

PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE

Separate RESET and ARM, two PNOZ stages, dual watchdog contacts, external adapters, redundant actuator interruption.

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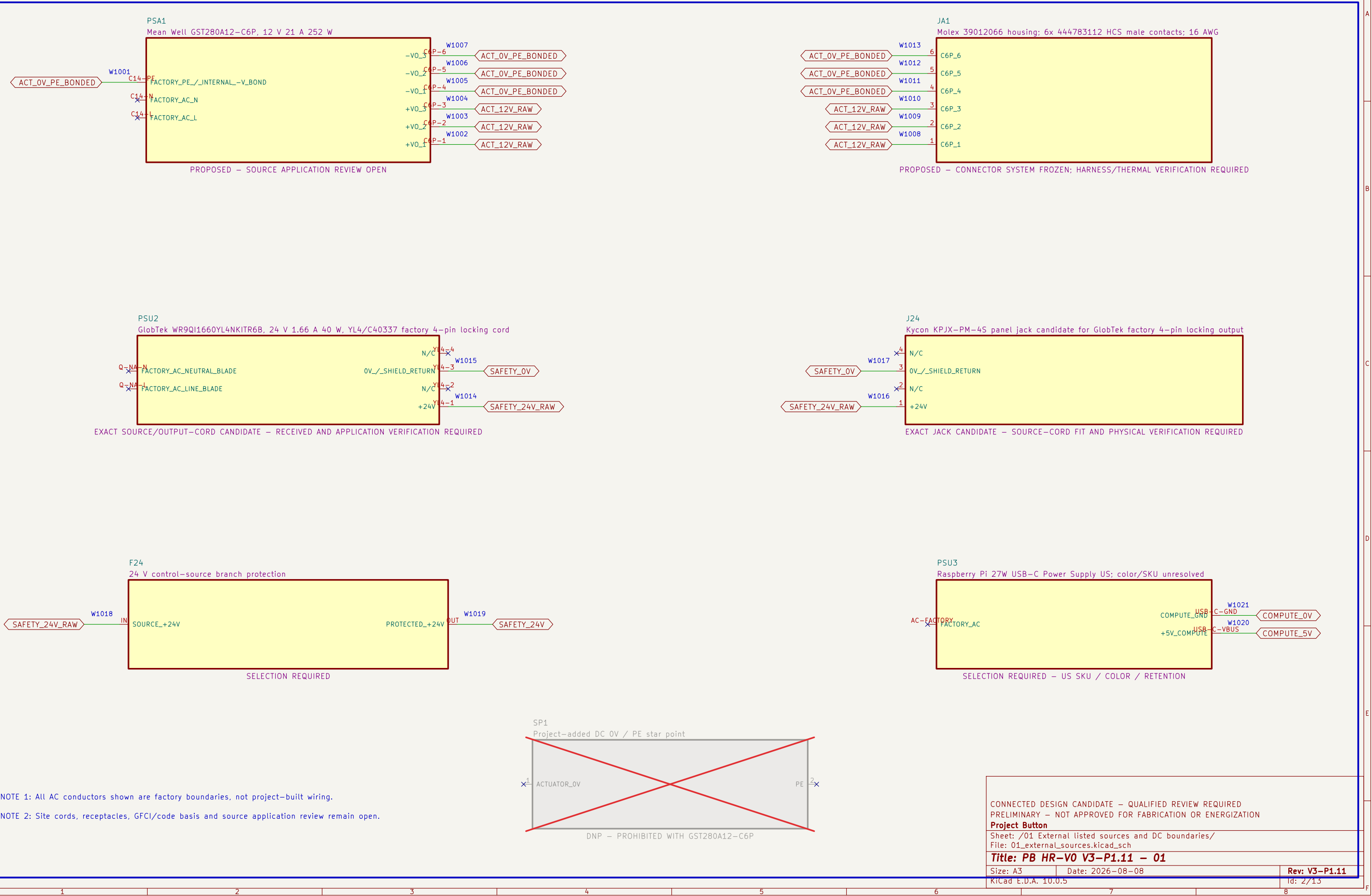
12 Watchdog PCB test access

File: 12_watchdog_pcb_test_access.kicad_sch

V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION Project Button		
Sheet: / File: project–button–v3.kicad_sch		
Title: PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE		
Size: A3	Date: 2026–08–08	Rev: V3–P1.11
KiCad E.D.A. 10.0.5		Id: 1/13

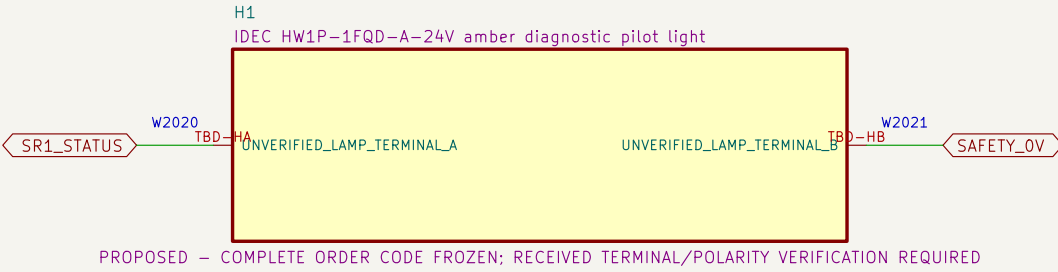
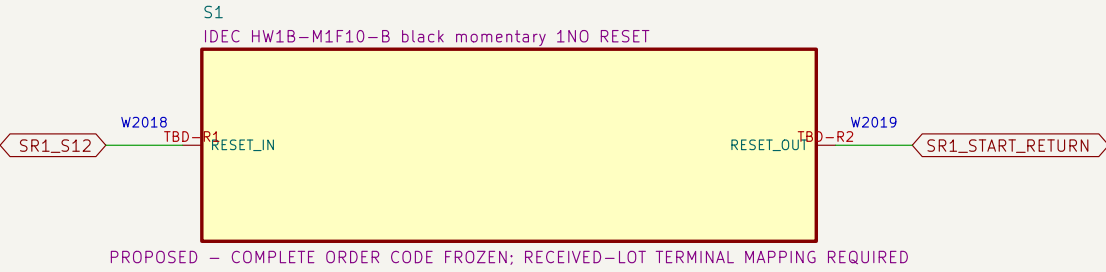
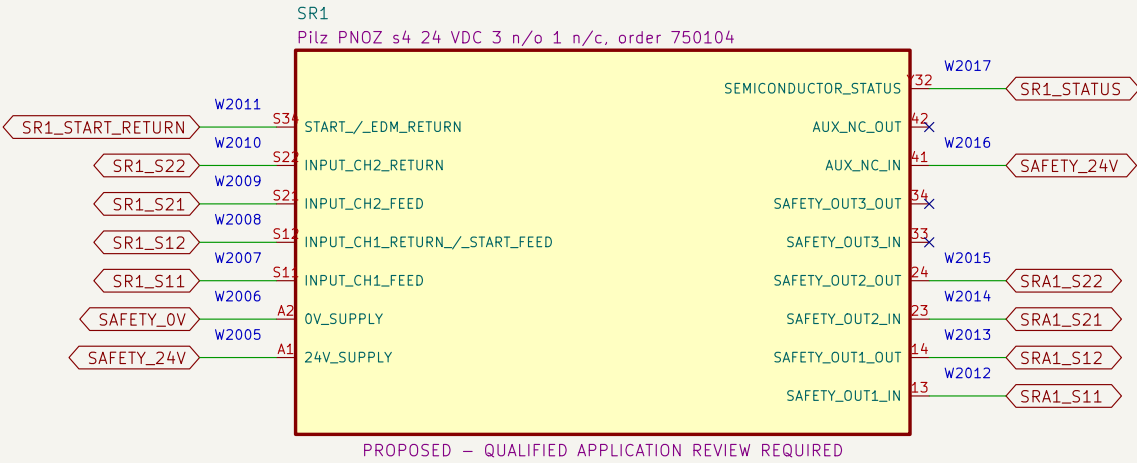
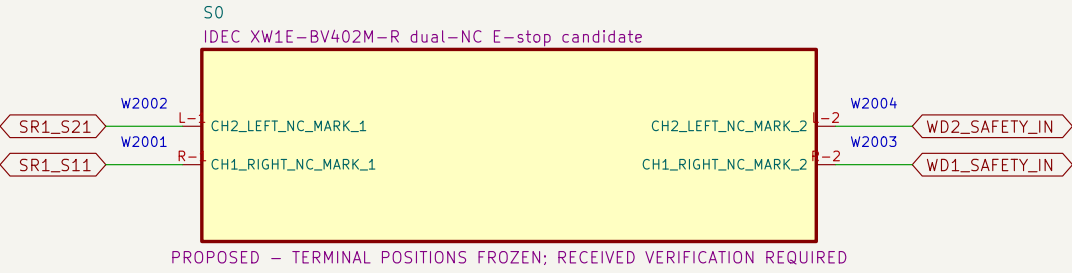
01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.



02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains one E-stop NC and one watchdog NO contact; RESET cannot energize K1/K2.



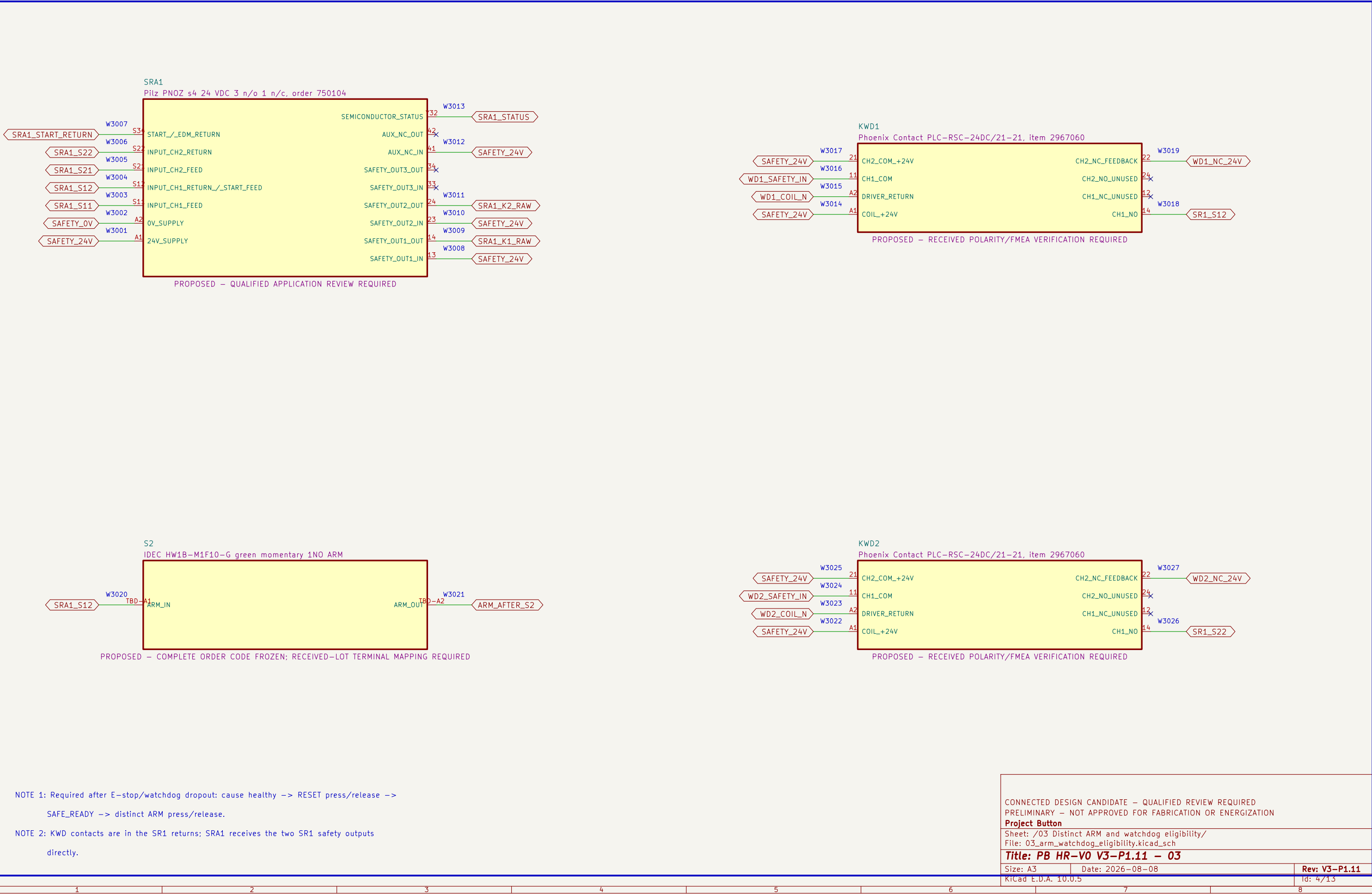
NOTE 1: Heartbeat loss opens both SR1 input returns; recovery alone cannot restore the monitored RESET stage.

NOTE 2: RESET release may make SR1 eligible, but SRA1 and K1/K2 remain de-energized until a later ARM.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /02 Dual-channel E-stop and RESET eligibility/ File: 02_estop_eligibility.kicad_sch		
Title: PB HR-V0 V3-P1.11 – 02		
Size: A3	Date: 2026-08-08	Rev: V3-P1.11
KiCad E.D.A. 10.0.5		Id: 3/13

03 Distinct ARM and watchdog eligibility

SRA1 requires SR1 eligibility, two watchdog channels, EDM proof and a new ARM action.



NOTE 1: SRA1 outputs are separately protected before K1 and K2 coils. FSR1/FSR2 use proposed Phoenix PT 4–HESI item 3211861 holders with D–ST 4 item 3030420 end cover; both fuse links, grouping and coordination remain SELECTION REQUIRED. NOTE 2: Loaded DC interruption, suppression behavior, mirror–contact use and coordination require qualified review.	CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION Project Button Sheet: /04 Contactor coils, mirror contacts and EDM/ File: 04_contactor_edm.kicad_sch Title: PB HR–V0 V3–P1.11 – 04 <table border="1"> <tr> <td>Size: A3</td><td>Date: 2026–08–08</td><td>Rev: V3–P1.11</td></tr> <tr> <td>KiCad E.D.A. 10.0.5</td><td></td><td>Id: 5/13</td></tr> </table>	Size: A3	Date: 2026–08–08	Rev: V3–P1.11	KiCad E.D.A. 10.0.5		Id: 5/13
Size: A3	Date: 2026–08–08	Rev: V3–P1.11					
KiCad E.D.A. 10.0.5		Id: 5/13					

links, grouping and coordination remain SELECTION REQUIRED.							Sheet: /04 Contactor coils, mirror contacts and EDM/ File: 04_contactor_edm.kicad_sch		
NOTE 2: Loaded DC interruption, suppression behavior, mirror-contact use and coordination							Title: PB HR-V0 V3-P1.11 - 04		
require qualified review.							Size: A3	Date: 2026-08-08	Rev: V3-P1.11
							KiCad E.D.A. 10.0.5	Id: 5/13	

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED	
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION	
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Size: A3	Rev: V3-P1.11
Date: 2026-08-08	Id: 5/13
KiCad E.D.A. 10.0.5	

Title: PB HR-V0 V3-P1.11 - 04		
Size: A3	Date: 2026-08-08	Rev: V3-P1.11
KiCad E.D.A. 10.0.5		Id: 5/13

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KiCad E.D.A. 10.0.5		Id: 5/13

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7	8	

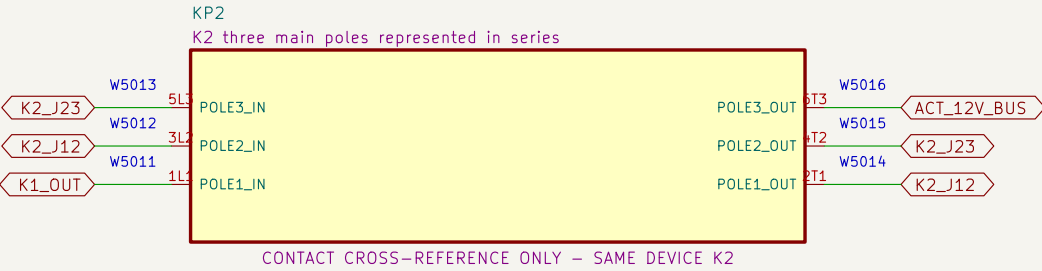
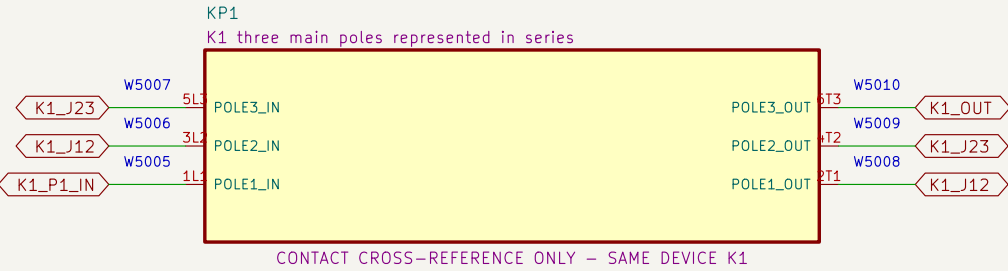
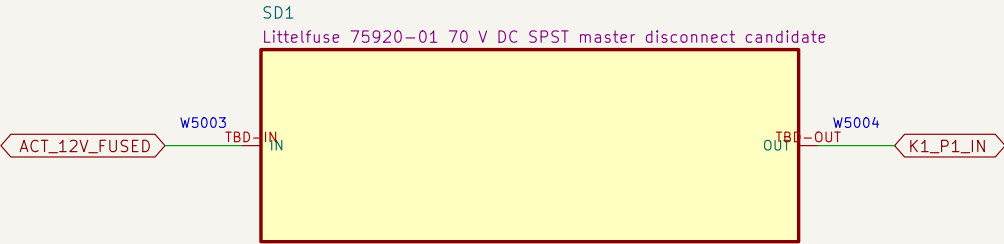
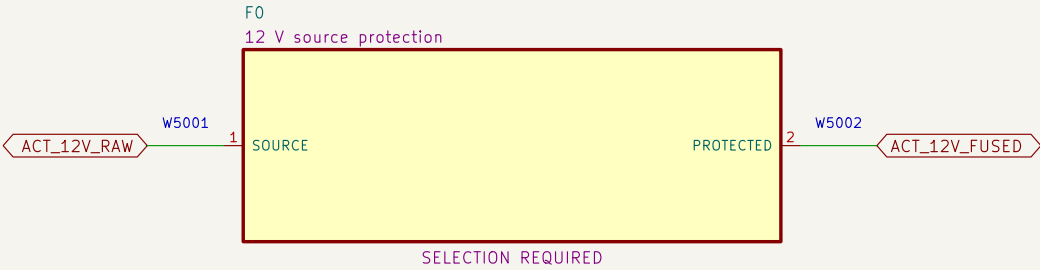
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	7	8	F
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05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



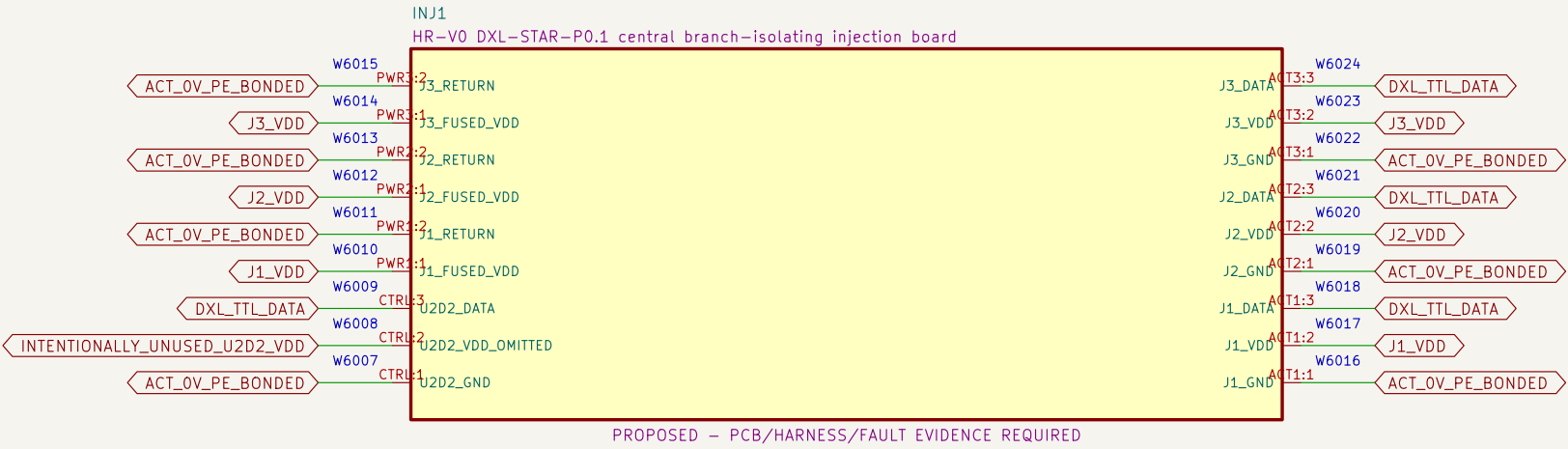
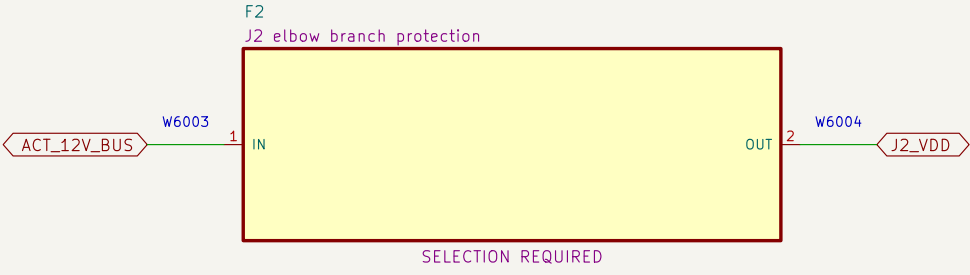
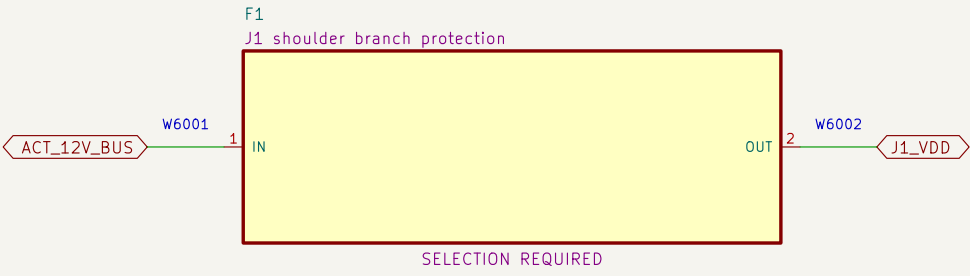
NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /05 Redundant actuator–power interruption/ File: 05_actuator_interruption.kicad_sch		
Title: PB HR–V0 V3–P1.11 – 05		
Size: A3	Date: 2026–08–08	Rev: V3–P1.11
KiCad E.D.A. 10.0.5		Id: 6/13

06 Protected actuator branches and central DYNAMIXEL star injection

Each actuator has a separate protected VDD branch; U2D2 pin 2 and inter-actuator VDD paths are omitted.



NOTE 1: INJ1 is a fixed central star; U2D2 pin 2 is unrouted and J1/J2/J3 positive rails never join.

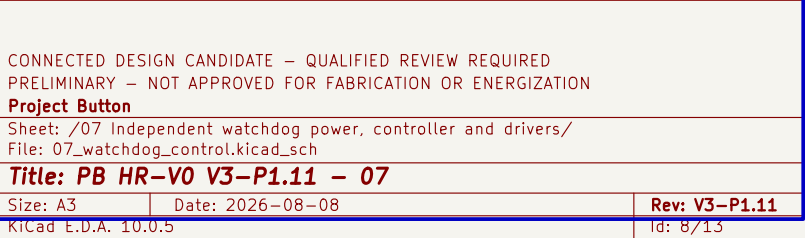
NOTE 2: Exact actuator cables, branch returns, enclosure, continuity, isolation, signal-integrity, pull and no-backfeed tests remain release gates.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button
Sheet: /06 Protected actuator branches and central DYNAMIXEL star injection/
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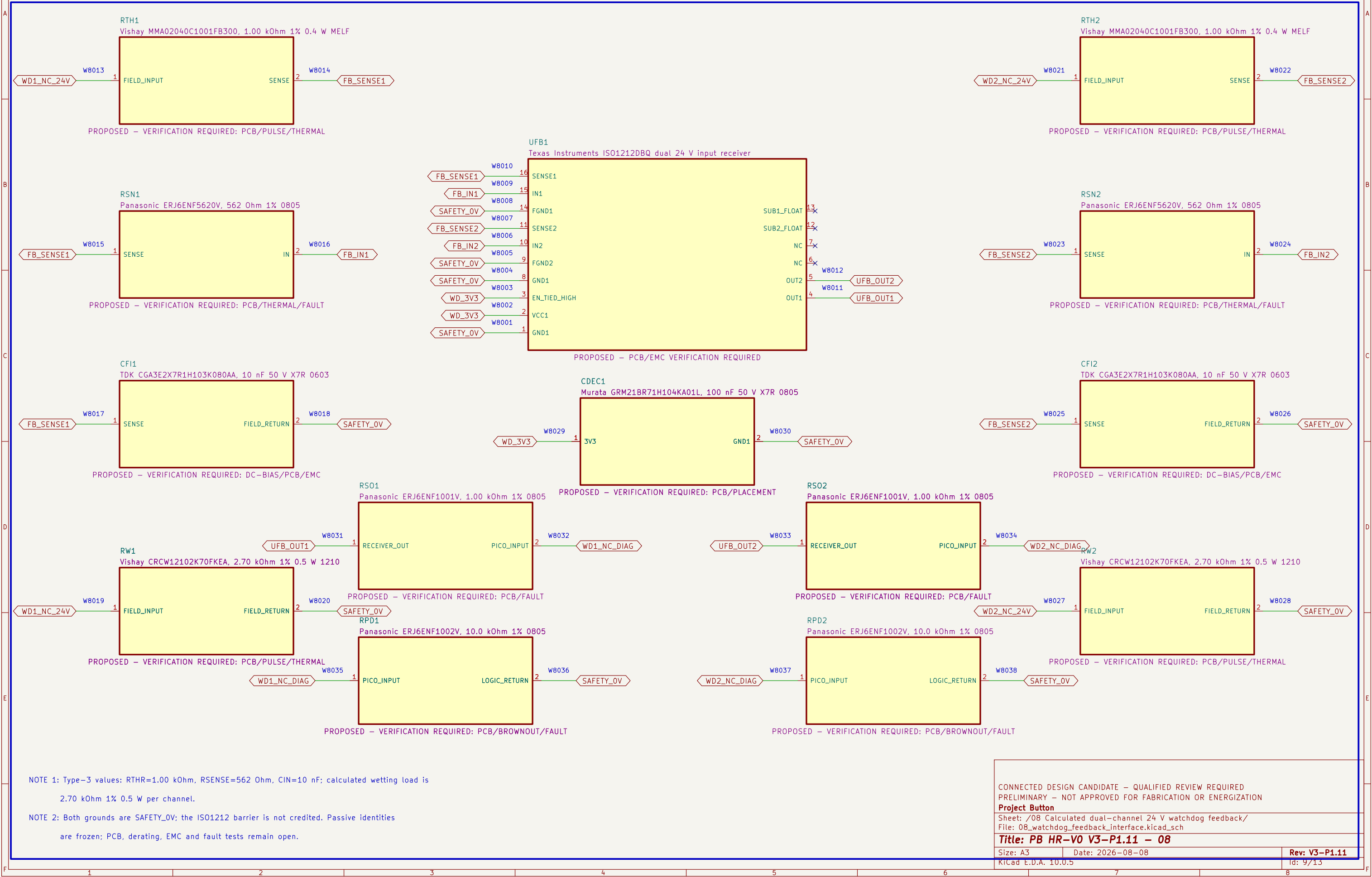
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Size: A3	Date: 2026-08-08	Rev: V3-P1.11
KiCad E.D.A. 10.0.5		Id: 7/13



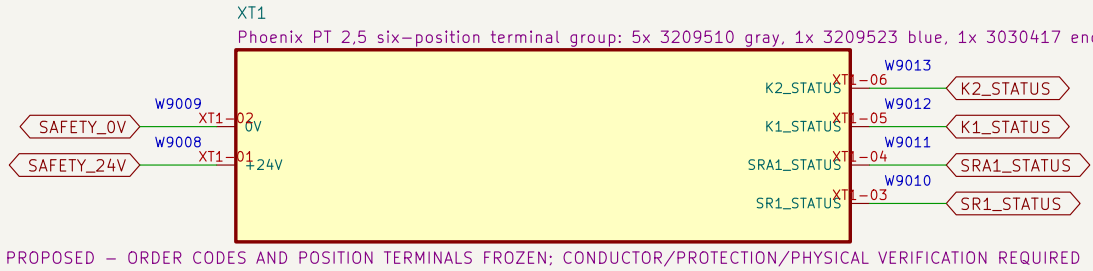
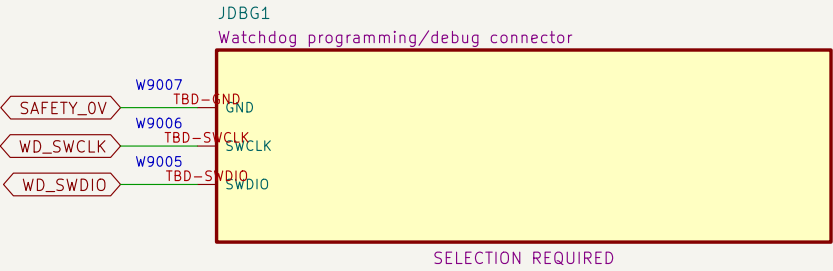
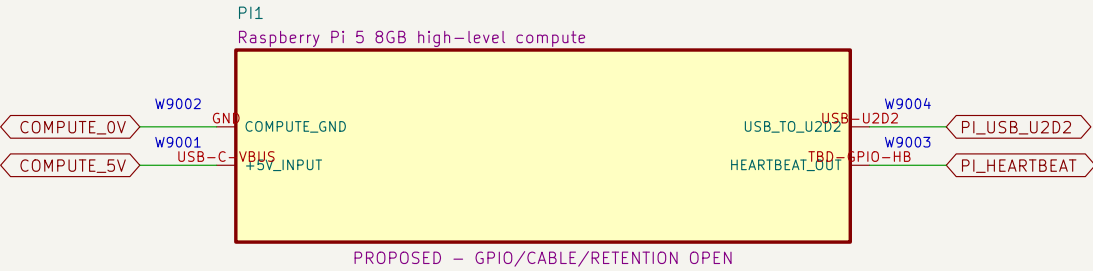
08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



09 Compute, debug and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



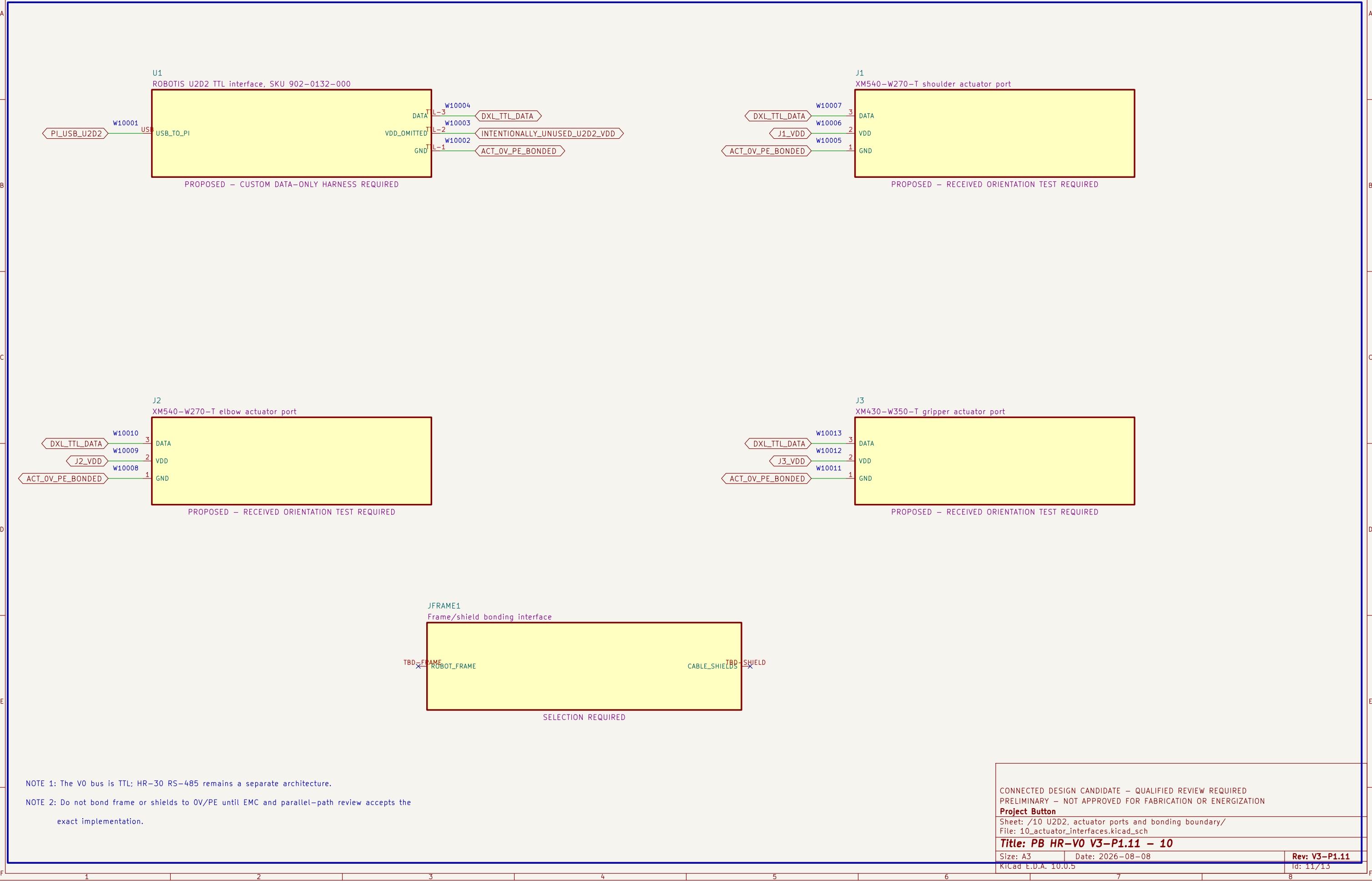
NOTE 1: Programming/debug connections must not enable outputs during operation.

NOTE 2: Terminal family, markers, conductor range, torque and enclosure layout remain selection required.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /09 Compute, debug and control terminals/ File: 09_compute_and_control_terminals.kicad_sch		
Title: PB HR-V0 V3-P1.11 – 09		
Size: A3	Date: 2026-08-08	Rev: V3-P1.11
KiCad E.D.A. 10.0.5	Id: 10/13	

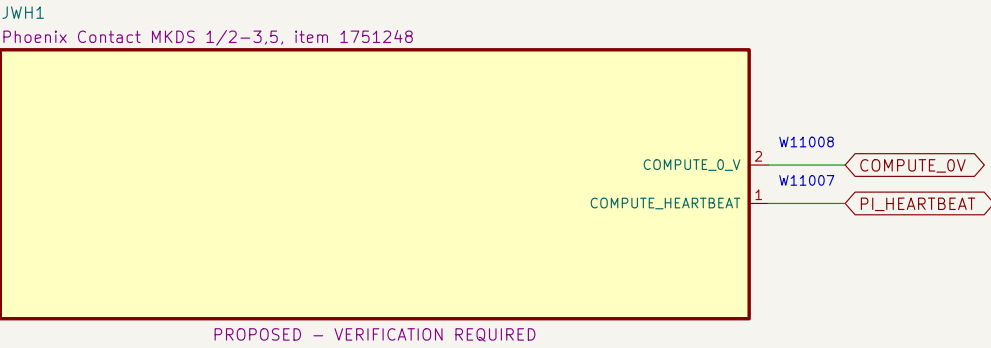
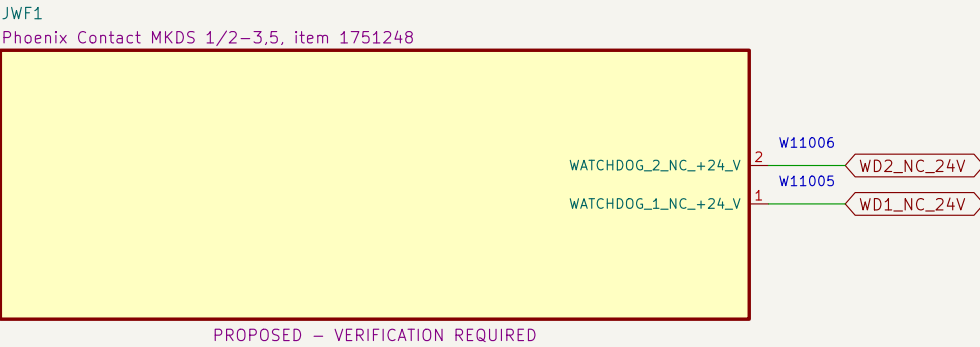
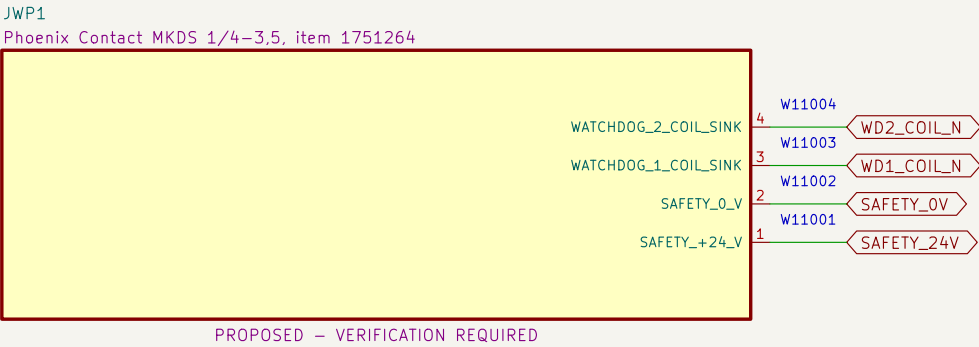
10 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.



11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.



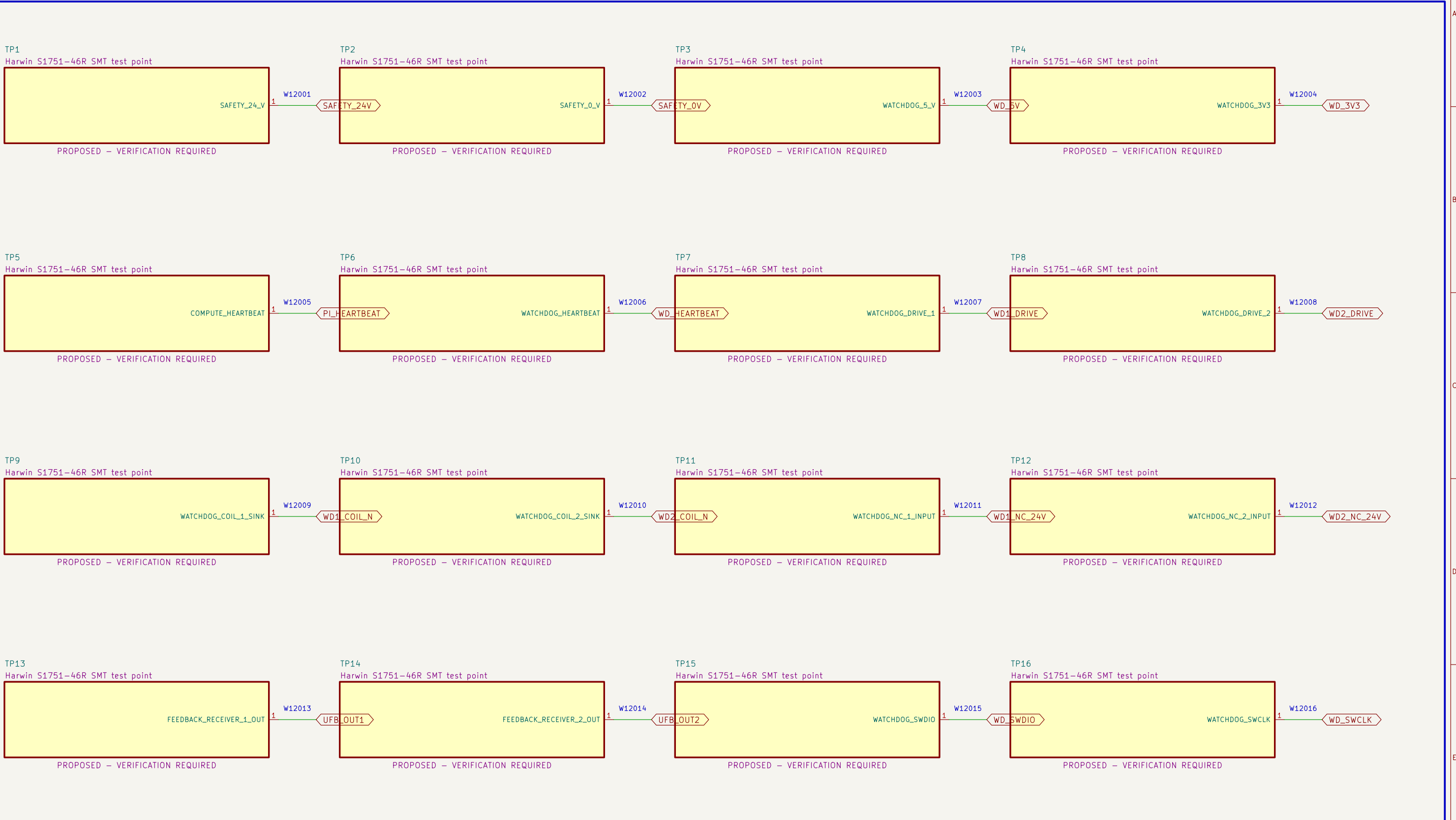
NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
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Size: A3	Date: 2026–08–08	Rev: V3–P1.11
KiCad E.D.A. 10.0.5		Id: 12/13

12 Watchdog PCB test access

Exact clip-compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /12 Watchdog PCB test access/		
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